

CHE 201 Fall 2019 Midterm

Majd Barakat

TOTAL POINTS

92 / 100

QUESTION 1

20 pts

1.1 a 8 / 10

- 0 pts Correct
- ✓ - 2 pts first one is wrong
- 2 pts second one is wrong
- 2 pts third one is wrong
- 2 pts fourth one is wrong
- 2 pts fifth one is wrong

1.2 b 5 / 5

- ✓ - 0 pts Correct
- 2 pts gage pressure is not converted to absolute pressure
- 2 pts T is not converted to K
- 5 pts Not done or completely wrong
- 3 pts only pressure is calculated

1.3 c 0 / 5

- 0 pts Correct
- ✓ - 5 pts incorrect

QUESTION 2

2 50 pts

2.1 a 3 / 3

- ✓ - 0 pts Correct

2.2 b 8 / 8

- ✓ - 0 pts Correct
- 1 pts PV state 2 is wrong
- 1 pts PV state 3 is wrong
- 1 pts TV state 2 is wrong
- 1 pts TV state 3 is wrong
- 8 pts incorrect

- 1 pts PV state 1 is wrong
- 1 pts TV state 1 is wrong
- 4 pts PV graph is wrong

2.3 c 5 / 5

- ✓ - 0 pts Correct
- 1 pts work expansion compression
- 2 pts KE and PE is zero
- 2 pts closed system is not listed

2.4 d 2 / 2

- ✓ - 0 pts Correct
- 1 pts delta E equation is not written
- 1 pts not simplified

2.5 e 15 / 15

- ✓ - 0 pts Correct
- 3 pts calculation error
- 3 pts v_3 is wrong
- 3 pts v_2 is wrong
- 15 pts incorrect
- 3 pts P is wrong
- 3 pts unit conversion
- 3 pts W_{12} is zero
- 2 pts not multiplied with mass

2.6 f 15 / 15

- ✓ - 0 pts Correct
- 3 pts u_3 is wrong
- 3 pts u_1 is wrong
- 2 pts not multiplied with mass
- 2 pts x value is wrong
- 2 pts calculation error
- 15 pts incorrect
- 3 pts q not calculated
- 3 pts not clear calculation

2.7 g 2 / 2

✓ - 0 pts Correct

- 0.5 pts M3v not calculated /wrong
- 0.5 pts m2v not calculated/ wrong
- 1 pts m1v not calculated / wrong

- 10 pts incorrect or not done

- 2 pts calculation error

QUESTION 3

3 30 pts

3.1 a 2 / 2

✓ - 0 pts Correct

- 2 pts not done
- 1 pts boundary not labeled

3.2 b 2 / 2

✓ - 0 pts Correct

- 0.5 pts Delta KE and PE is zero not listed
- 1 pts ideal gas is not listed
- 1 pts closed system is not listed

3.3 C 2 / 3

- 0 pts Correct

✓ - 1 pts Z is not listed or wrong

- 3 pts not done
- 2 pts delta E equation is not listed
- 1 pts not simplified

3.4 d 3 / 3

✓ - 0 pts Correct

- 3 pts wrong
- 1 pts slope needs to be negative
- 1 pts Not curved, plotted as linear

3.5 e 10 / 10

✓ - 0 pts Correct

- 10 pts not correct or not done
- 6 pts error with integration
- 4 pts not clear explanation
- 2 pts calculation error

3.6 f 10 / 10

✓ - 0 pts Correct

NAME: Mey D BarakatPlease fill in answers in the space supplied and show all your work to receive full credit.
Useful Tables at end of Exam.1. a) (10 pts) State whether the following statements are **True or False**

T An ideal gas at a given state expands to a fixed final volume at constant pressure or at constant temperature. Work is greater in constant temperature case.

F from the graph we notice area is greater and - constant temp
A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality of 0.5.

F Because quality depends of mass it is a mass fraction
As pressure decreases from the critical pressure, the values of the specific volume of saturated vapor and saturated liquid approach each other.

F as pressure decreases, they get further away from the top of dome and they SL and SV don't approach
If a system undergoes a polytropic process, the system must consist of an ideal gas.

T It takes more energy to vaporize 1kg of saturated liquid water at 100°C than it would at 120°C. because the dome becomes narrower and thus less energy work is needed to cross the dome

b) (5 pts) The gage pressure of an automobile tire is measured to be 210kPa before a trip and 220kPa after the trip at a location where the atmospheric pressure is 95kPa.

Assuming the volume of the tire remains constant and the air temperature before the trip is 25°C, determine the air temperature in the tire after the trip. (Air is ideal gas)

$P_{\text{gage}} = 210 \text{ kPa}$ before, $P_{\text{gage}} = 220 \text{ kPa}$ after, $P_{\text{atm}} = 95 \text{ kPa}$, $V = \text{constant}$, $T_1 = 25^\circ\text{C}$

$P_{\text{absolute}} = P_{\text{gage}} + P_{\text{atm}} = 305 \text{ kPa}$, same calc for $P_{\text{absolute}} = 315 \text{ kPa}$ after

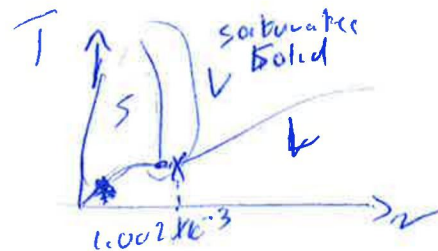
since V and n are constant
 $\frac{P_1 V}{T_1} = \frac{P_2 V}{T_2} \Rightarrow T_2 = T_1 \times \frac{P_2}{P_1} = \frac{(25 + 273.15) \times 315 \text{ kPa}}{305 \text{ kPa}} = 341.78^\circ\text{C}$

c) (5pts) The specific volume of saturated liquid water at 0.01°C is $1.002 \times 10^{-3} \text{ m}^3/\text{kg}$.

What would be the specific volume of saturated solid at the same temperature, greater or less than saturated liquid value? Why?

looking at the graph we notice that v values are smaller

than the point (X) therefore the saturated solid v are smaller



NAME: May D Bantut

CHE 201 MidTerm Exam 1

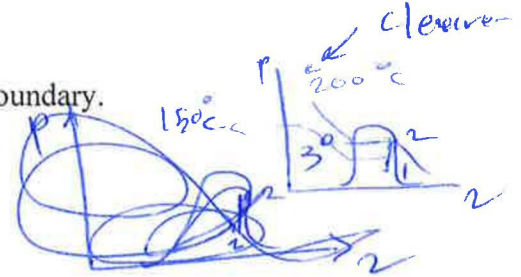
October 11th, 2019

2. (50 pts) 2kg of liquid-vapor mixture of water contained in a piston-cylinder assembly undergoes two processes in series from an initial state where the temperature is 120°C.

- **Process 1-2** System is heated at constant volume to a saturated vapor state at 200°C.
- **Process 2-3** System is compressed at constant pressure until it reaches to 150°C.

Kinetic and potential effects can be ignored.

- (3 pts) Draw a schematic of the system and label the system boundary.
- (8 pts) Carefully sketch the process on a:
 - (1) P-v diagram (Pressure-specific volume)
 - (2) T-v diagram (Temperature-specific volume)
- (5 pts) State the engineering model
- (2 pts) Write the simplified energy balance
- (15 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?
- (15 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?
- (2 pts) Calculate the mass of vapor in each state in kg.



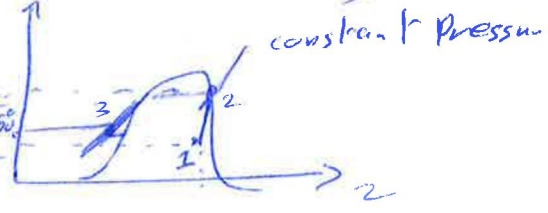
system is water inside piston-cylinder

Engineering model:
 • system is closed
 • $\Delta KE = \Delta PE = 0$
 • work is expansion/contraction

$m = 2 \text{ kg}, T_1 = 120^\circ\text{C}$

$\Delta E = \Delta KE + \Delta PE + \Delta U = Q - W$

$\Rightarrow \Delta U = Q - W$



State 2: $v_1 = v_2, x = 1, T_2 = 200^\circ\text{C} \Rightarrow v_1 = v_2 = 0.1274 \text{ m}^3/\text{kg}$

$W_{1 \rightarrow 2} = 0$ constant volume $\rightarrow P_2 = 15.54 \text{ bar}$

$P_2 = P_3 = 15.54 \text{ bar}$

state 3 is in the ~~super~~ compressed liquid region
 we can approximate it to the saturated liquid value of the same temperature (150°C) with a low % error

$\Rightarrow v_3 = 1.0905 \times 10^{-3} \text{ m}^3/\text{kg}$

$W_{2 \rightarrow 3} = \int_{v_2}^{v_3} P dv = m P_{2/3} (v_3 - v_2) = 2 \text{ kg} (15.54 \text{ bar} \times 10^5 \text{ N/m}^2) (1.0905 \times 10^{-3} \text{ m}^3/\text{kg} - 0.1274 \text{ m}^3/\text{kg})$

$= -392569.5 = -392.57 \text{ kJ}$ work is done on the sys.

$$W_{\text{total}} = W_{1 \rightarrow 2} + W_{2 \rightarrow 3} = \underline{\underline{-392.57 \text{ kJ}}}$$

work is done on the sys'

we need to find the quality of state 1

$$T = 120^\circ\text{C} \quad v_1 = 0.1274 \text{ m}^3/\text{kg} = (1-x)v_f + x v_g \leftarrow 0.8919 \text{ m}^3/\text{kg}$$

$\swarrow 1.0603 \times 10^{-3} \text{ m}^3/\text{kg}$

$$\Rightarrow x = 0.142$$

$$U_1 = (1 - 0.142) 503.50 \text{ kJ/kg} + (0.142) (2529.3 \text{ kJ/kg}) = 791.16 \text{ kJ/kg}$$

we do the same thing we did to get v_3 to get U_3
at ~~120~~ 150°C, the saturated liquid U_f is 631.68 kJ/kg

$$\Rightarrow m \Delta U = m(U_3 - U_1) = Q - W$$

$$\Rightarrow 2 \text{ kg} (631.68 - 791.16) \text{ kJ/kg} = Q - (-392.57 \text{ kJ})$$

$$\Rightarrow Q = -318.96 \text{ kJ} - 392.57 \text{ kJ}$$

= -711.53 heat is transferred from the sys

$$m \text{ of } V \text{ in state 2} = \frac{2 \text{ kg} \cdot 0.1274 \text{ m}^3/\text{kg}}{0.1274 \text{ m}^3/\text{kg}} = \underline{\underline{2 \text{ kg}}}$$

$$m \text{ vapor in state 1} = (0.142) 2 \text{ kg} = 0.284 \text{ kg}$$

$m \text{ vapor in state 3} = 0$ all of it is compressed liquid

NAME: Majid Bando

CHE 201 MidTerm Exam 1

October 11th, 20193. (30 points) A piston cylinder device initially contains 0.4 m^3 of air at 100 kPa and 240°C .

The air is now compressed to 0.1 m^3 in such a way that the temperature inside the cylinder remains constant. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (2 pts) State the engineering model
- (3 pts) Write the simplified energy balance and compressibility factor (Z).
- (3 pts) Draw P-v diagram (Pressure-specific volume)
- (10 pts) Determine the work (kJ)
- (10 pts) Determine the heat (kJ)

$$V_1 = 0.4 \text{ m}^3, P_1 = 100 \text{ kPa}, T_1 = 240^\circ \text{C}$$

$$V_2 = 0.1 \text{ m}^3, T_1 = T_2 = 240^\circ \text{C}$$

system is air inside piston cylinder

engineering model:

• System is closed

$$\Delta \text{KE} = \Delta \text{PE} = 0$$

• work is expansion/contraction

• Assume ideal gas

$$\Delta E = \Delta \text{KE} + \Delta \text{PE} + \Delta U = Q - W \Rightarrow \Delta U = Q - W$$

$$T \text{ is constant} \Rightarrow P V = \text{constant}$$

$$\Rightarrow P_2 V_2 = R T_2$$

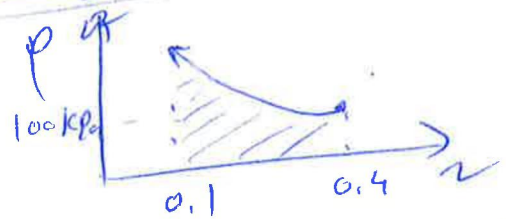
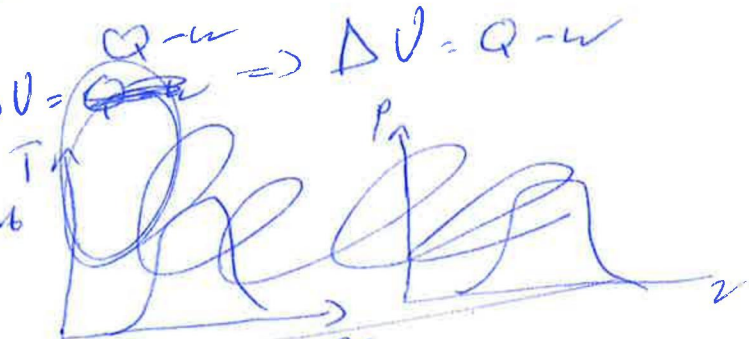
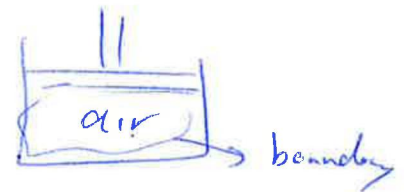
$$\Rightarrow P_2 = \frac{R T_2}{V_2} = \frac{(8.314 \text{ kJ/kmol} \cdot \text{K})}{0.1 \text{ m}^3} (240^\circ \text{C} + 273.15 \text{ K})$$

$$P_1 V_1 = \text{constant} = 100 \text{ kPa} \times 0.4 \text{ m}^3 = 40000 \text{ kPa} \cdot \text{m}^3$$

$$P_2 = \frac{P_1 V_1}{V_2} = \frac{(100 \text{ kPa})(0.4 \text{ m}^3)}{0.1 \text{ m}^3} = 400 \text{ kPa}$$

$$W = \int_{V_1}^{V_2} P dV = \int_{V_1}^{V_2} \frac{40000}{V} dV = -55451.77 \text{ J} = -55.45 \text{ kJ}$$

on the system



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CHE 201 MidTerm Exam 1

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at 240°C

~~At~~ $U_1 = 171.13 \text{ kJ/kg} = U_2$

$$\Rightarrow \Delta U = 0 = Q - W \Rightarrow 0 = Q - (-55.45 \text{ kJ})$$

$$\Rightarrow \underline{\underline{Q = -55.45 \text{ kJ}} \text{ from the sys}}$$

